CANDIDATE – PLEASE NOTE!
PRINT your name below and return this booklet with the answer sheet. Failure to do so may result in disqualification.

FORM TP 2017050

TEST CODE **01212010**

MAY/JUNE 2017

CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN SECONDARY EDUCATION CERTIFICATE® EXAMINATION

CHEMISTRY

Paper 01 – General Proficiency

1 hour 15 minutes

09 JUNE 2017 (p.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This test consists of 60 items. You will have 1 hour and 15 minutes to answer them.

2. In addition to this test booklet, you should have an answer sheet.

3. Each item in this test has four suggested answers lettered (A), (B), (C), (D). Read each item you are about to answer and decide which choice is best.

4. On your answer sheet, find the number which corresponds to your item and shade the space having the same letter as the answer you have chosen. Look at the sample item below.

   Sample Item

   The SI unit of length is the

   (A) metre
   (B) newton
   (C) second
   (D) kilogram

   Sample Answer

   ● (B) (C) (D)

   The best answer to this item is "metre", so (A) has been shaded.

5. If you want to change your answer, erase it completely before you fill in your new choice.

6. When you are told to begin, turn the page and work as quickly and as carefully as you can. If you cannot answer an item, go on to the next one. You may return to that item later.

7. You may do any rough work in this booklet.

8. Figures are not necessarily drawn to scale.

9. You may use a silent, non-programmable calculator to answer items.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

1. Which of the following processes provide evidence of the particulate nature of matter?

   I. Diffusion
   II. Filtration
   III. Osmosis

   (A) I and II only
   (B) I and III only
   (C) II and III only
   (D) I, II and III

Item 2 refers to the following table which gives the melting points and boiling points of the chlorides of four elements.

Element	Melting Point of Chloride (K)	Boiling Point of Chloride (K)
I	463	453
II	248	348
III	973	1693
IV	193	333

2. Which of the elements forms a chloride that is ionic in nature?

   (A) I
   (B) II
   (C) III
   (D) IV

3. Which of the following elements has 7 electrons in its outer shell?

   (A) Hydrogen
   (B) Oxygen
   (C) Nitrogen
   (D) Chlorine

4. An atom has 20 electrons. When it ionizes, it has the same electronic configuration as

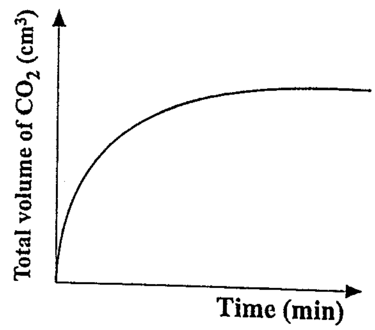
   (A) neon
   (B) argon
   (C) sodium
   (D) calcium

5. Which of the following elements does NOT form simple ions by gaining or losing electrons?

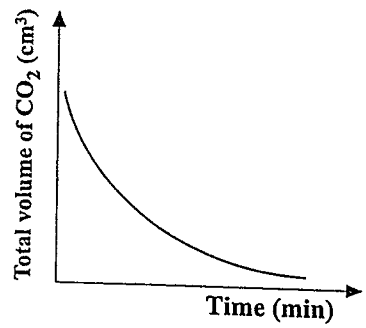
   (A) Carbon
   (B) Copper
   (C) Calcium
   (D) Chlorine

6. Some calcium carbonate was reacted with excess dilute hydrochloric acid. The volume of carbon dioxide evolved was recorded and plotted against time. Which of the following graphs represents the reaction?

(A)



(B)



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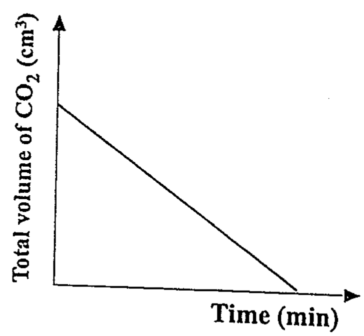
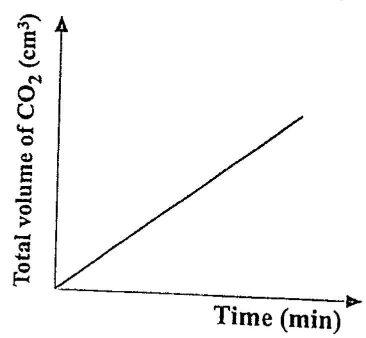


(D)



7. Which of the following represents a balanced equation for the reaction between aqueous silver nitrate and aqueous sodium chloride?

(A) $Ag(NO_3)_2(aq) + 2NaCl(aq) \rightarrow AgCl_2(s) + 2NaNO_3(aq)$

(B) $AgNO_3(aq) + 2NaCl(aq) \rightarrow AgCl_2(s) + Na_2NO_3(aq)$

(C) $Ag_2NO_3(aq) + NaCl(aq) \rightarrow Ag_2Cl(s) + NaNO_3(aq)$

(D) $AgNO_3(aq) + NaCl(aq) \rightarrow AgCl(s) + NaNO_3(aq)$

8. Which of the following acids will NOT form an acid salt?

(A) H_3PO_4
(B) H_2CO_3
(C) H_2SO_4
(D) CH_3CO_2H

9. Sodium reacts with water according to the equation

$$2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g).$$

(Molar volume of gas = 24 litres at rtp)

The number of litres of hydrogen (at rtp) liberated when 0.1 mole of sodium reacts with excess water is

(A) 1.2
(B) 2.4
(C) 12.0
(D) 24.0

10. Which of the following oxides may show both acidic and basic properties?

(A) Iron(II) oxide
(B) Sodium oxide
(C) Calcium oxide
(D) Aluminium oxide

11. A mixture of copper(II) oxide and copper(II) sulfate is BEST separated by

(A) distilling the mixture
(B) heating the mixture and condensing
(C) shaking with excess water and then filtering
(D) shaking with excess water followed by fractional distillation

12. Which of the following elements exhibits allotropy?

(A) Silicon
(B) Carbon
(C) Uranium
(D) Aluminium

13. Which of the following statements is TRUE of an endothermic reaction?

(A) Heat is given up to the surroundings.
(B) Heat is absorbed from the surroundings.
(C) The products have less energy than the reactants.
(D) The products have the same energy as the reactants.

14. In which of the following substances will electrolysis NOT occur when an electrical current is passed through it?

(A) Sodium hydroxide solution
(B) Molten sodium chloride
(C) Solid sodium chloride
(D) Dilute sulfuric acid

15. Which of the following are physical properties of ionic substances?

I. High melting points
II. Soluble in organic solvents
III. Good conductors of electricity in the liquid state

(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III

16. Which of the following substances, when added to pure water, SIGNIFICANTLY increases the water's conductivity?

(A) Graphite
(B) Iron(II) hydroxide
(C) Copper(II) hydroxide
(D) Sodium chloride

17. A separating funnel can be used to separate a mixture of

(A) water and ethanol
(B) water and kerosene
(C) water and solid sodium chloride
(D) kerosene and solid sodium chloride

Items **18–19** refer to the following terms.

(A) Ionic crystals
(B) Simple molecular
(C) Macromolecular
(D) Metallic

In answering Items 18–19, each term may be used once, more than once or not at all.

18. Which of the above terms describes the structure of sodium chloride?

19. Which of the above terms describes the structure of copper?

20. The pH of fresh sugar cane juice, which is usually 5.0–5.5, can be changed to 7.5–8.0 for more efficient processing by adding

(A) acetic (ethanoic) acid, CH_3CO_2H
(B) sodium chloride, $NaCl$
(C) slaked lime, $Ca(OH)_2$
(D) limestone, $CaCO_3$

21. Which TWO of the following statements describe acid salts?

I. Their anions cannot dissociate to give hydrogen ions.
II. They form a normal salt when all replaceable hydrogen of the acid is removed.
III. Their anions can dissociate to give hydrogen ions.
IV. They are not capable of generating a normal salt.

(A) I and III
(B) I and IV
(C) II and III
(D) III and IV

22. A piece of metal is reacted with an acid to produce hydrogen gas. Which of the following procedures should be employed in order to increase the rate of the reaction?

I. Increasing the temperature at which the reaction is carried out
II. Subdividing the lump of metal
III. Reducing the concentration of the acid

(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III

23. The atomic number of an element is defined as the number of

(A) electrons
(B) protons
(C) protons and neutrons
(D) electrons and neutrons

24. Which of the following halogens is a liquid at room temperature?

(A) Bromine
(B) Fluorine
(C) Chlorine
(D) Iodine

25. Which of the following statements is TRUE for the equation below?

$$Mg(s) + H_2SO_4(aq) \rightarrow MgSO_4(aq) + H_2(g)$$

(A) Hydrogen is oxidized from 0 to +1.
(B) Hydrogen is reduced from +2 to 0.
(C) Mg is oxidized from 0 to +2.
(D) Mg is reduced from +2 to 0.

26. The element that is used to determine the relative atomic mass of other elements is

(A) Carbon-10
(B) Carbon-12
(C) Carbon-13
(D) Carbon-14

Items 27–28 refer to the following table showing various tests for gases.

	Test	Observation
(A)	Damp blue litmus paper	Damp blue litmus turns red
(B)	Damp red litmus paper	Damp red litmus turns blue
(C)	Glowing splint	Splint relights
(D)	Lit splint	Splint extinguishes with a pop

In answering Items **27–28**, each option may be used once, more than once or not at all.

27. Which test can be used to identify an acidic gas?

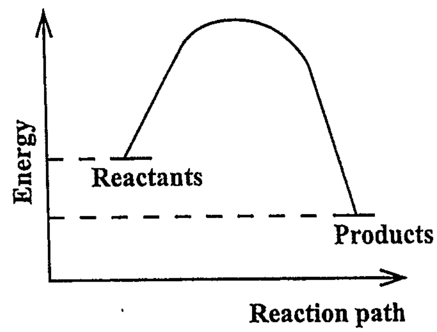
28. Which test can be used to identify ammonia gas?

29. Which of the following does NOT take place during electrolysis?

(A) Oxidation occurs at the anode.
(B) Anions move towards the cathode.
(C) Cations move towards the cathode.
(D) Inert electrodes do not react with the product.

30. Which of the following diagrams illustrates the course of an exothermic reaction?

(A)



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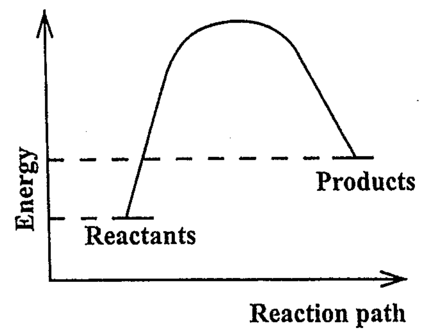

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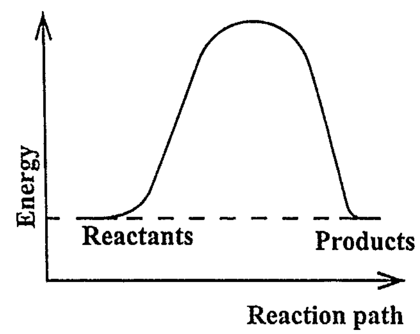
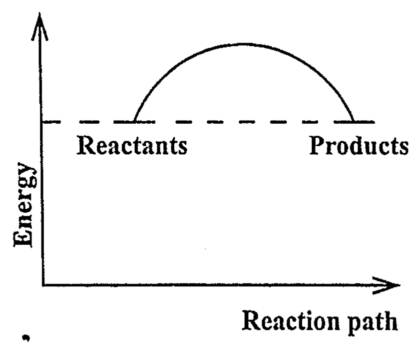


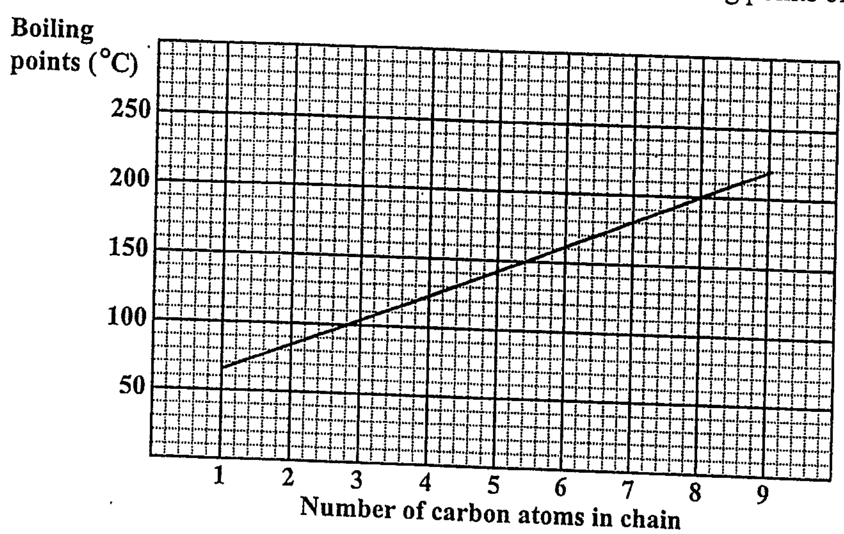
(D)



31. An atom with atomic structure $^{27}_{13}X$ contains

(A) 14 protons, 13 neutrons
(B) 13 protons, 27 neutrons
(C) 13 protons, 14 neutrons
(D) 27 protons, 13 neutrons

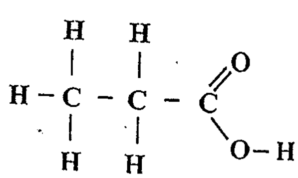
Item 32 refers to the following graph which shows the boiling points of some straight-chain alcohols.



32. Which of the following molecular formulae represents the alcohol whose boiling point is 180 °C?

(A) $CH_3(CH_2)_3OH$
(B) $CH_3(CH_2)_4OH$
(C) $CH_3(CH_2)_5OH$
(D) $CH_3(CH_2)_6OH$

33. To which homologous series does the following structure belong?



(A) Carboxylic acids
(B) Alkanes
(C) Alkenes
(D) Alcohols

34. Which of the following is the name of the compound $CH_3CH_2CH_2CH_2OH$?

(A) 1-propanol
(B) 4-propanol
(C) 4-butanol
(D) 1-butanol

Items 35–36 refer to the following conversions of ethanol.

(A) $CH_3CH_2OH \rightarrow CH_3COOH$
(B) $CH_3CH_2OH \rightarrow CH_3CHO$
(C) $CH_3CH_2OH \rightarrow C_2H_4$
(D) $CH_3CH_2OH \rightarrow 2CO_2 + 3H_2O$

In answering Items 35–36, each conversion may be used once, more than once or not at all.

35. Which of the conversions above correctly represents the dehydration of ethanol?

36. Which of the conversions above requires manganate(VII)/$KMnO_4$?

Items 37–38 refer to the following equation.

Organic Acid + Alcohol $\rightleftharpoons$ Ester + Water.

37. The forward reaction is known as

(A) synthesis
(B) hydrolysis
(C) esterification
(D) polymerization

38. The reverse reaction is known as

(A) synthesis
(B) hydrolysis
(C) esterification
(D) polymerization

39. 'Cracking' is the process of breaking down

(A) large alkane molecules into smaller alkene and alkane molecules
(B) large alkane molecules into smaller alkane and hydrogen molecules
(C) complex sugar molecules into simple sugar molecules
(D) large polymer molecules into smaller polymer molecules

Item 40 refers to the following equation.

$RCOOH(aq) + NaOH(aq) \rightarrow RCOONa(aq) + H_2O(l)$

40. Which of the following processes correctly describes the equation above?

(A) Decomposition
(B) Neutralization
(C) Precipitation
(D) Reduction

41. An organic substance, when reacted with bromine water, decolourizes it immediately. The substance is MOST likely an

(A) alcohol
(B) alkane
(C) alkene
(D) ester

42. Glucose is converted to starch or cellulose by

(A) dehydrogenation
(B) oxidation and reduction
(C) addition polymerization
(D) condensation polymerization

43. Compounds in a homologous series

(A) exist in the same state
(B) possess the same molecular formula
(C) differ successively by a $-CH_2$ group
(D) possess only carbon in straight chains

44. Ethene reacts with hydrogen, chlorine and hydrogen chloride. These reactions are all examples of

(A) addition
(B) substitution
(C) condensation
(D) polymerization

45. Which of the following substances is NOT a polymer?

(A) Nylon
(B) Protein
(C) Fructose
(D) Cellulose

46. Which of the following methods is used for the extraction of aluminium?

(A) Electrolysis of its molten oxide
(B) Electrolysis of its aqueous chloride
(C) Reduction of its oxide using coke
(D) Reduction of its oxide using carbon monoxide

47. Which of the following statements about sulfur are true?

I. It is used in the vulcanization of rubber.
II. It is used in the extraction of iron from iron ore.
III. It is used in the manufacture of explosives.

(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III

48. Soapless detergents are

(A) made from fats
(B) all biodegradable
(C) inorganic materials
(D) not affected by hard water

49. A solution, when treated with sulfur dioxide gas, becomes green. The solution contains potassium

(A) nitrate
(B) sulfate
(C) dichromate(VI)
(D) manganate(VII)

50. A suitable drying agent for the laboratory preparation of ammonia gas is

(A) calcium oxide
(B) calcium chloride
(C) anhydrous copper sulfate
(D) concentrated sulfuric acid

51. When copper(II) carbonate is heated alone in a dry test tube, a gas is evolved and a black residue is formed. This gas is expected to

(A) turn red litmus blue
(B) relight a glowing splint
(C) decolourize acidified aqueous potassium manganate(VII)
(D) form a white precipitate with aqueous calcium hydroxide

52. An element that is a nonmetal

(A) is a solid at room temperature
(B) can act as a reducing agent
(C) forms an acidic oxide
(D) conducts electricity

53. Which of the following observations is likely when EXCESS aqueous ammonia is added to a solution of copper(II) sulfate and the mixture is shaken?

(A) Green solution
(B) Green precipitate
(C) Deep blue solution
(D) Light blue precipitate

Items 54–55 refer to the following metals.

(A) Al
(B) Cu
(C) Fe
(D) Na

In answering Items **54–55**, a particular metal may be used once, more than once or not at all.

54. Which metal reacts vigorously with water to produce a strongly alkaline solution?

55. Which metal has an alloy which is used in the manufacture of aircraft?

56. Which of the following reactions is LEAST likely to occur?

(A) $Mg(s) + Zn(NO_3)_2(aq) \rightarrow Mg(NO_3)_2(aq) + Zn(s)$

(B) $Cu(s) + 2AgNO_3(aq) \rightarrow Cu(NO_3)_2(aq) + 2Ag(s)$

(C) $Fe(s) + Cu(NO_3)_2(aq) \rightarrow Fe(NO_3)_2(aq) + Cu(s)$

(D) $Pb(s) + Fe(NO_3)_2(aq) \rightarrow Pb(NO_3)_2(aq) + Fe(s)$

57. The main substances responsible for 'acid rain' are

(A) chlorofluorocarbons
(B) lead compounds in exhaust fumes
(C) carbon dioxide and carbon monoxide
(D) sulfur dioxide and sulfur trioxide

58. Which of the following observations is expected when aqueous silver nitrate is added to aqueous potassium chloride?

(A) A pungent gas is evolved.
(B) A blue precipitate appears.
(C) A white precipitate appears.
(D) A brown precipitate appears.

59. Which metal appears between copper and aluminium in the reactivity series and is an important component in haemoglobin?

(A) Lead
(B) Iron
(C) Zinc
(D) Magnesium

60. Which of the following gases will give a positive test with dry cobalt chloride paper?

(A) O_2
(B) H_2
(C) CO_2
(D) H_2O

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.